

Orientation and Onboarding Plan

Learner Profile

The learners are third-year medical students completing a specialty 3-week clerkship in Hepatology. The group size for this activity is 2 students.

- **Level of Training:** These students are transitioning from hybrid lecture-based preclinical learning to practical clinical applications of knowledge. They are expected to have a good understanding of basic anatomy and physiology but lack hands-on procedural exposure experience.
- **Prior Knowledge and Experience:** Students possess a fundamental understanding of liver anatomy and portal hypertension. However, there is a commonly held misconception in patients with chronic liver disease that an elevated international normalized ratio INR on serum testing leads to an increased bleeding risk which is contrary to patients without liver disease. While they may have been taught about paracentesis through lectures, they likely possess minimal hands-on experience.
- **Educational Considerations:** As adult learners, these students benefit from structured, psychologically safe environments where mistakes can be converted into learning opportunities. Because learners are expected to have diverse learning preferences (visual, tactile, and auditory), the orientation and simulation will incorporate mixed-modality training, combining didactic elements with hands-on practice.

Orientation Goals

1. **Cognitive:** Before the paracentesis simulation, the learner will recall the core components of the Serum-Ascites Albumin Gradient and list at least 2 different etiologies for ascites.
2. **Psychomotor:** Prior to paracentesis simulation, the learner will identify the basic equipment and surface anatomy landmarks necessary to conduct a paracentesis after being provided with a standardized procedural checklist.
3. **Affective:** Before the simulation begins, the learner will recognize the paracentesis safety parameters as identified by national society guidelines allowing them to feel prepared in their role as an evidence-based advocate.

Key Onboarding Components

The following components will be provided on the day of the simulation.

Clinical environment and expectations. Students will receive a brief verbal overview of the simulation setting, the standardized patient or mannequin model being used, and the faculty observer role. The structured observation checklist used to assess Goals 1 and 2 will be shared with students in advance so expectations are transparent.

Policies, protocols, and safety considerations. Students will be provided with a one-page reference summarizing current practice guidance recommendations, including the importance of universal inpatient paracentesis and a brief review of indications and contraindications (O’Leary et al., 2019).

Procedural resources. Students will be oriented to the simulation equipment and given access to a stepwise procedural checklist that mirrors the assessment tool used for Goal 2. They will have five to ten minutes to review it before the case begins.

Roles and responsibilities. Students will individually complete the simulated paracentesis with 1:1 supervision by the faculty member, and each will have the same role (i.e., medical resident). Each student will be observed individually on the affective and psychomotor objectives. The cognitive objective will be assessed collectively during the post-procedure debrief.

Timeline and Delivery

All onboarding occurs on the day of the simulation during a single 20-minute orientation block. During these 20 minutes, faculty will introduce the simulation objectives, the structured checklist, and the case framing. This will be followed by students independently reviewing the one-page guideline reference and procedural checklist. Finally, there will be a facilitated discussion to clarify questions about equipment, roles, and expectations before the case begins.

The paracentesis simulation will then take place, approximately 15 minutes per student (i.e., 30 minutes) followed by a 10-minute post-simulation debrief. Therefore, the entire activity will take 1 hour.

Connection to Learning Objectives

This orientation and onboarding plan is designed to build on the foundational knowledge students have previously obtained while addressing all three learning objectives. Sharing the guideline reference in advance ensures students have a fair opportunity to recognize the INR-based deferral as incorrect (Objective 1), while the procedural checklist preview allows for a safe learning environment and clinical confidence (Objectives 2 and 3).

References

1. Biggins SW, Angeli P, Garcia-Tsao G, et al. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology*. 2021;74(2):1014-1048. <https://doi.org/10.1002/hep.31884>
2. O'Leary, J. G., Greenberg, C. S., Patton, H. M., & Caldwell, S. H. (2019). AGA Clinical Practice Update: Coagulation in Cirrhosis. *Gastroenterology*, 157(1), 34–43.e1. <https://doi.org/10.1053/j.gastro.2019.03.070>